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Unit – 3 : Session – 12 : SLO – 1 | Circular Queue — Dequeue Operation

■ I. Algorithm to Delete an Element from a Circular Queue

- Step 1: IF FRONT = -1
- Write UNDERFLOW
- Goto Step 4
- [END OF IF]
- Step 2: SET VAL = QUEUE[FRONT]
- Step 3: IF FRONT = REAR
- SET FRONT = REAR = -1
- ELSE
- IF FRONT = MAX-1
- SET FRONT = 0
- ELSE
- SET FRONT = FRONT + 1
- [END OF IF]
- [END OF IF]
- Step 4: EXIT

■ II. Programming Code — delete_element()

```
int delete_element() {  
    int val;  
    if (front == -1) {  
        printf("\n UNDERFLOW");  
        return -1;  
    }  
    val = queue[front];  
    if (front == rear)  
        front = rear = -1;  
    else {  
        if (front == MAX-1)  
            front = 0;  
        else  
            front++;  
    }  
    return val;  
}
```